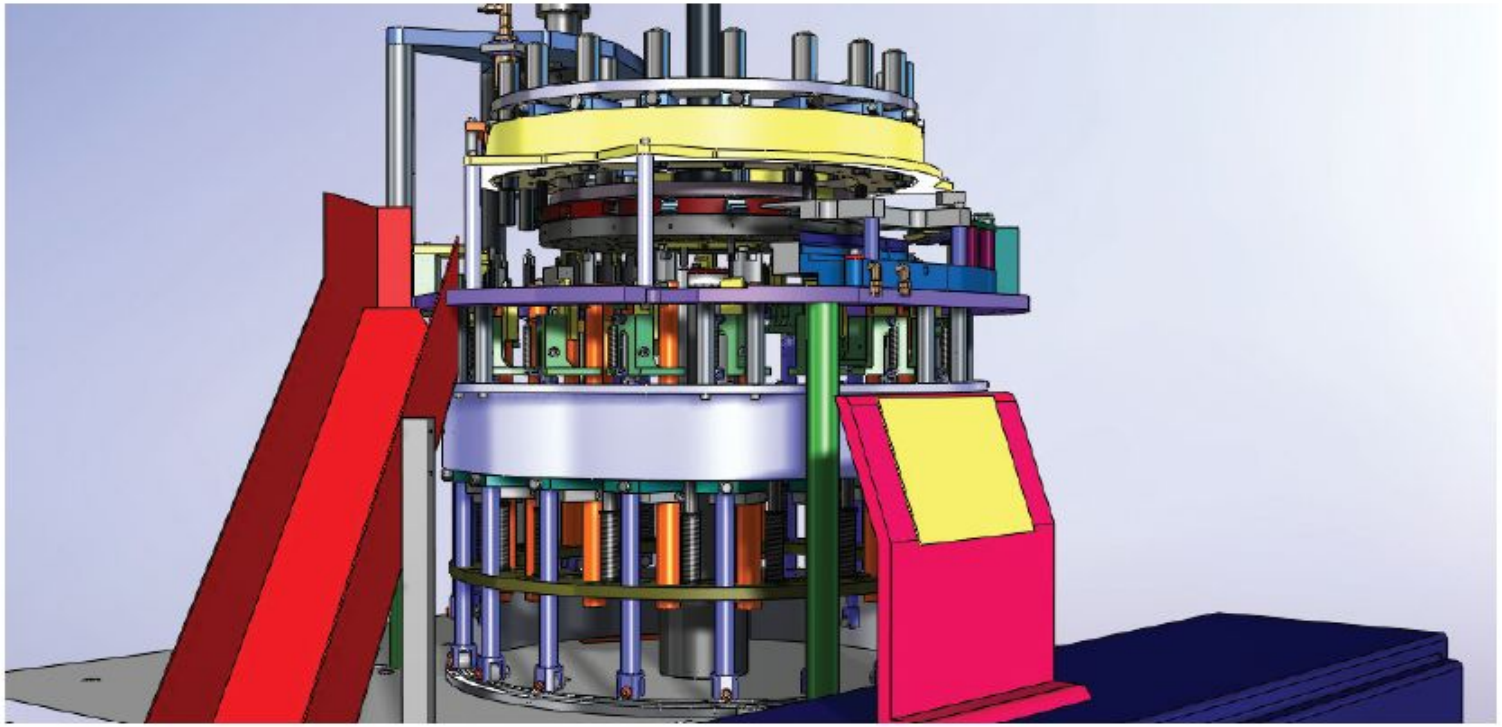
Bridging the Worlds of 2D and 3D CAD Design



Learning Objective

1 **Know the advantages of using CAD in comparison with other methods**

Advantages of CAD: quality; accuracy; time; cost; electronic transfer of information; links with other software eg CAD/CAM, rendering software, animation software, finite element analysis (FEA)

Other methods: manual drafting; model making

Learning Outcomes

You should be able to:

- **ALL** – students be able to state and explain ***three*** benefits of 3D CAD models compared to 2D drawings.
- **MOST** – students be able to state and explain ***Five*** benefits of 3D CAD models compared to 2D drawings.
- **SOME** – students be able to state and explain ***eight*** benefits of 3D CAD models compared to 2D drawings. And justify why they are important to a manufacturing organisation.

Overview

As manufacturers rapidly change from 2D to 3D CAD in today's digital world, designers are demanding 3D to ***enhance*** their *designs and improve communication*. From *websites to rapid prototyping - customers, designers and engineers now depend on the latest 3D modelling tools*.

This presentation highlights the ***advantages*** of designing with SolidWorks ***3D CAD*** compared to ***2D CAD*** methods

Advantages of 3D CAD

As you work with 3D CAD software, you will quickly discover that 3D ***enhances*** not only the ***design process***, but also ***communication*** with *customers and design teams*.

Advantages of 3D CAD

The ***main advantages*** of using SolidWorks ***3D CAD software*** *(or other 3D CAD software)* are:

1. *Enhance visualisation and communication*
2. *Eliminate manual updates*
3. *Reduce errors with interference and collision checking*
4. *Reuse existing designs*
5. *Accelerate development cycles with virtual testing and optimisation*
6. *Design for manufacturing*
7. *Create bills of materials and manage data smoothly*
8. *Sales and marketing tools*

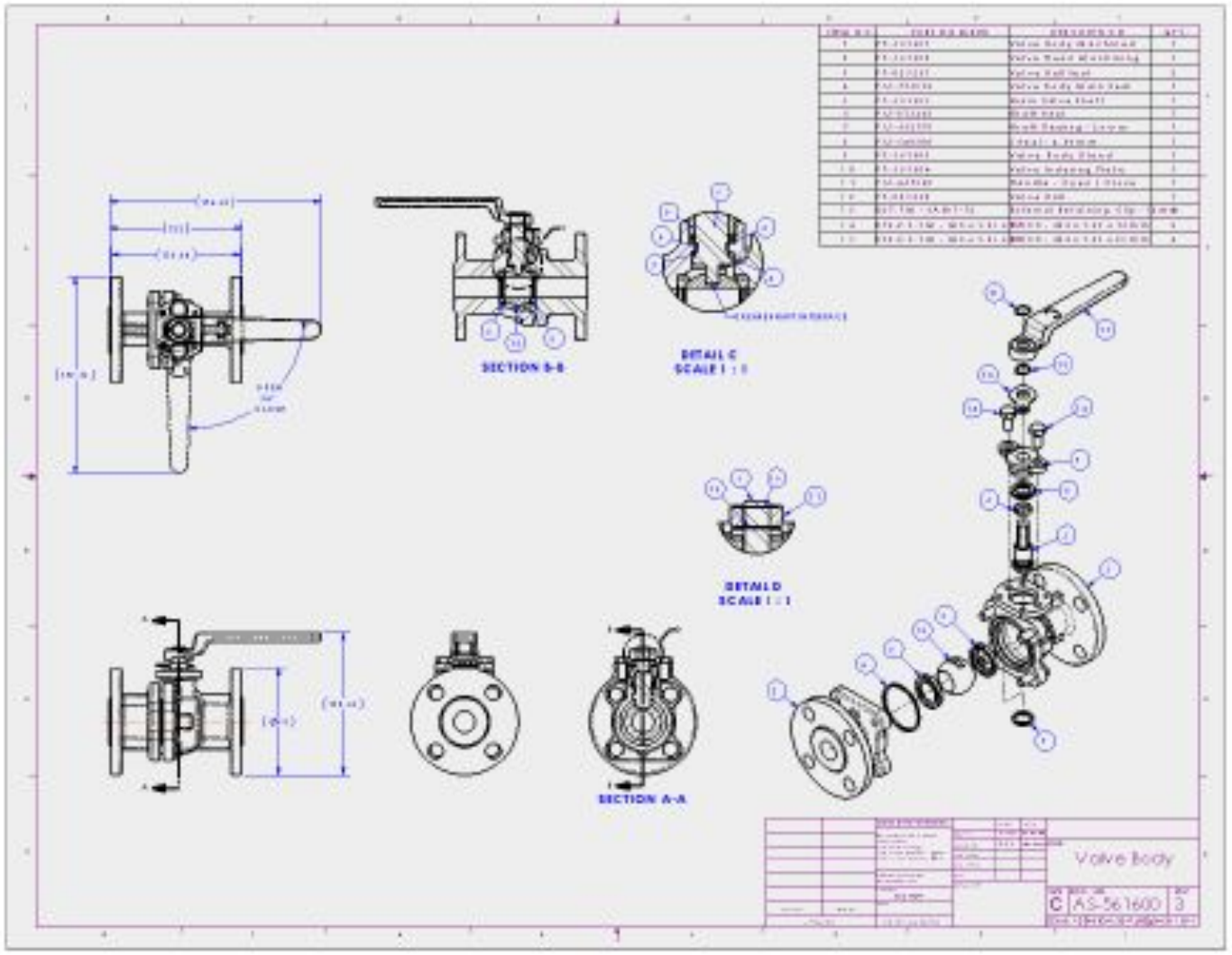


With SolidWorks Professional software, Multiquip has improved efficiencies in virtually every downstream function in the product development process.

1. Enhance visualisation and communication

Enhance visualisation and communication

CAD is all about ***communicating ideas***. We live in a 3D world, so we visualise objects in the same way. *When it comes to communicating a design, we naturally prefer a 3D image, model, or animation over a 2D technical drawing. In the 2D world, designers must be able to look at three or four views of a design and mentally combine them in order to visualise what that design will look like in 3D.*


SECTION B-B
DETAIL C
SCALE 1 : 1
DETAIL D
SCALE 1 : 1
SECTION A-A
Valve Body
C
AS-561600
3

Enhance visualisation and communication Cont.

While engineers and drafters can understand a 2D drawing - *customers, salespeople, buyers, and suppliers* may find it much more difficult to comprehend.

Looking at a design in 3D versus 2D eliminates the need for viewers to have mastered this technical knowledge.



By moving from 2D to SolidWorks 3D design, HSG discovered how improved visualization fosters collaboration and innovation.

Case Study:



Rick Morse

Pearce Processing Systems in Gloucester, Massachusetts

Rick Morse, owner of *Pearce Processing Systems in Gloucester, Massachusetts*, says *that being able to* ***present his proposals*** *and* ***finished designs*** *in 3D to his customers makes* ***communication*** *much easier than providing them 2D drawings.*

Morse also makes *extensive use of* ***3D animation.***

The ability to show his customers a ***"movie"*** of the machine in action not only *communicates* how the machine will *function*, but also gives him an ***edge*** over others who are submitting 2D drawings in the quoting phase.

With ***3D images*** *and* ***3D animations*** potential customers feel more confident in what ***Morse*** will be delivering as a finished product.

Combine this with the ability to add ***photorealistic finishes*** ***(Rendered Images)*** to 3D models, and you can often find it difficult to tell if you are looking at a CAD model or a photograph of the actual physical machine.

Enhance visualisation and communication Cont.

In addition to presentation benefits, ***3D assembly models*** can be quickly and easily ***"exploded"*** to create ***exploded views*** for *technical illustrations* and *assembly instructions* without requiring any further drafting. And when 2D drawings are needed for manufacturing, 3D CAD provides ***automatic view creation (orthographic views)*** for any 2D view needed, including ***section views*** and ***detailed views***.

Enhance visualisation and communication Cont.

The ability to generate *3D images* and *animations* also makes it easier to *communicate* with others besides *customers*. *Sales, marketing, field service, operations staff, financial personnel, and management* may also find it challenging to interpret a 2D drawing. Yet they will readily understand the design when presented in 3D where you can ***rotate, zoom, measure, animate, and even "walk through"*** *your designs*.



3D CAD software allows you to thoroughly explore an assembly with explode, unexplode, and hide parts from views capabilities.

Summary:

Enhance visualisation and communication

1. *We naturally prefer a 3D image over a 2D technical drawing*
2. *Easy for non-engineers to understand*
3. *Can produce a 3D Assembly of product being designed*
4. *Applying photorealistic finishes (Rendered Images)* - difficult to tell if you are looking at a CAD model or a photograph
5. *Can produce a 3D Exploded model for technical illustrations* and *assembly instructions*
6. *Producing a 'movie' or animation product will give you a competitive edge*
7. *Can automatically generate 2D orthographic drawings*

Task 1

3D CAD models can be used to ***enhance*** the *visualisation and communication* of an engineering design.

1. ***State three*** ways a ***3D CAD model*** can be used to enhance an engineering design. For example – the designer/drafter can produce ***photorealistic (Rendered) images*** of the product.
2. ***Explain*** how your answers from ***question (1)*** enhance an engineering design.
3. How does ***Morse*** think these methods of enhancement ***benefit*** his ***design/manufacturing organisation***???

2. Eliminate manual updates

Eliminate manual updates

In 2D, you have to ***manually*** *update every drawing view whenever a* ***change*** *occurs*.

Change ***one part*** *and you not only have to include that change in* ***each*** *of the* ***three*** *drawing views for the part, you must also change* ***every view*** *of* ***every assembly*** *in which that part is used.*

Case Study:



Rick Morse

Pearce Processing Systems in Gloucester, Massachusetts

Rick Morse says he wonders *how they ever made a machine in the past in **2D CAD**. He spent much of his design time just trying to keep all his drawing views up to date as his designs evolved.*

Morse *relates that in designing complex food processing machines, his company incorporates many design changes for both machine* ***function*** *and* ***manufacturing cost*** *reasons*.

*Because he had to manually control the coordination between hundreds of drawing views in 2D CAD, there were thousands of opportunities for **mistakes**.*

*So, eliminating the need to **interpret** the design is one of the ways that working in 3D **saves time**.*

Eliminate manual updates Cont.

Consider how many views need to be updated manually in 2D each time a simple change is made to your design.

A ***minor*** change to a ***dimension*** on a part triggers a series of updates.

First, consider the drawing of the part - all the views (at least ***three*** views) must be modified.

Eliminate manual updates Cont.

Then, drawings of the ***assemblies*** that contain ***that part*** - again, most likely ***three*** *views*, ***must*** be ***updated***. And:

- What happens if that part exists multiple times in an assembly?
- Also, how can you be sure you updated all the drawings in which that part is used?

Eliminate manual updates Cont.

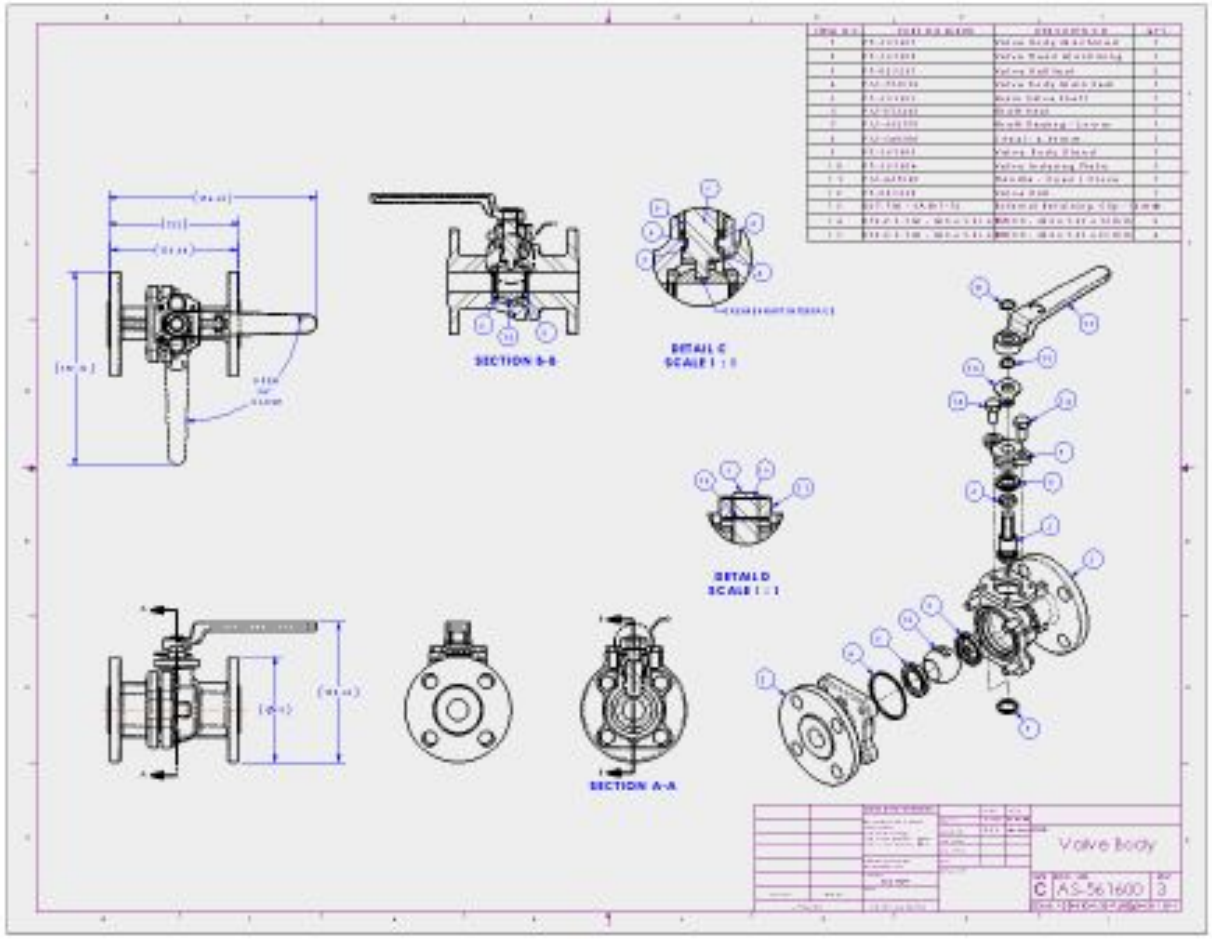
Luckily in SolidWorks software, you don't have to worry about any of this. Introducing the concept of ***associativity***.

In SolidWorks software, when you change a ***part model***, such as the ***length*** of the *part or* ***hole diameter***, or even ***add*** a ***new feature*** to a part, the change is ***automatically*** *rippled through to every drawing view, every assembly, and anywhere else that part is used*.

Eliminate manual updates Cont.



And when you want to know what other files will be affected by the change, SolidWorks software also provides the ability to ***automatically*** *track and identify where the part is used - what subassembly, what higher level assembly, and what drawings*, so that you can make sure you are modifying only files and designs that you really want to modify.



2D drawing views are automatically created from the 3D model and updated whenever the model is changed.

Summary:

Eliminate manual updates

1. *2D CAD - have to **manually** update every drawing view whenever a **change** occurs*
2. *2D CAD - design time spent trying to keep drawing views up to date - thousands of opportunities for **mistakes***
3. *Eliminating the need to **interpret** the design is one of the ways that working in 3D **saves time***
4. If a change is made to a 3D model - the change is ***automatically** rippled through to every drawing view, every assembly, and anywhere else that part is used*
5. *Using 3D software the designer has the ability to **automatically** track and identify where the part is used*

Task 2

3D CAD models can be used to ***eliminate*** the manual updating/revising of 2D drawings.

1. ***Explain*** how 3D CAD models eliminate the problem experienced when updating 2D CAD drawings.
2. Why is it a problem for a designer/drafter to modify a part ***drawn*** in 2D???
3. How does ***Morse*** think drawings which automatically update when a 3D CAD model is changed ***benefits*** *his* ***design/ manufacturing organisation***???



3. Reduce errors with interference and collision checking

Reduce errors with interference and collision checking

On a *2D drawing*, ***part interferences*** are difficult to find, *especially when the design becomes large and complicated*. In addition, because updates to 2D take so much time, many users often take shortcuts, *like changing a dimension on a part without updating the actual size of the model.*

*The phrase **"the drawing is not to scale"** is used a lot when dealing with 2D drawings.*

Reduce errors with interference and collision checking Cont.

Add to this the fact that *multiple designers* will be sharing the *assembly design duties*, and the ***potential*** for ***interfering parts*** almost becomes a *certainty*.

Checking 2D drawings to identify possible ***interference issues*** is ***extremely time-consuming***, and *interferences inevitably fall through the cracks, even with the most diligent checkers*.

Reduce errors with interference and collision checking Cont.

Using the SolidWorks software, you can *eliminate **interference** between parts*. *Interference* checking is ***automatic***, and ***every part*** can be checked to see if it ***interferes*** with any other part. ***Interference problems*** are ***highlighted***, and even the amount of the interference is reported. *With SolidWorks software you can also automate the process of **hole alignment** checking*.

Reduce errors with interference and collision checking Cont.

*Checking interference in an assembly that is static is difficult enough in **2D**, but it becomes almost impossible when you are dealing with a design that **moves**, like a packaging machine, or a piece of automation equipment. There are virtually an infinite number of possibilities for a **collision** to occur in designs that move. In **2D** there is really **no** practical way to check for a **collision**.*

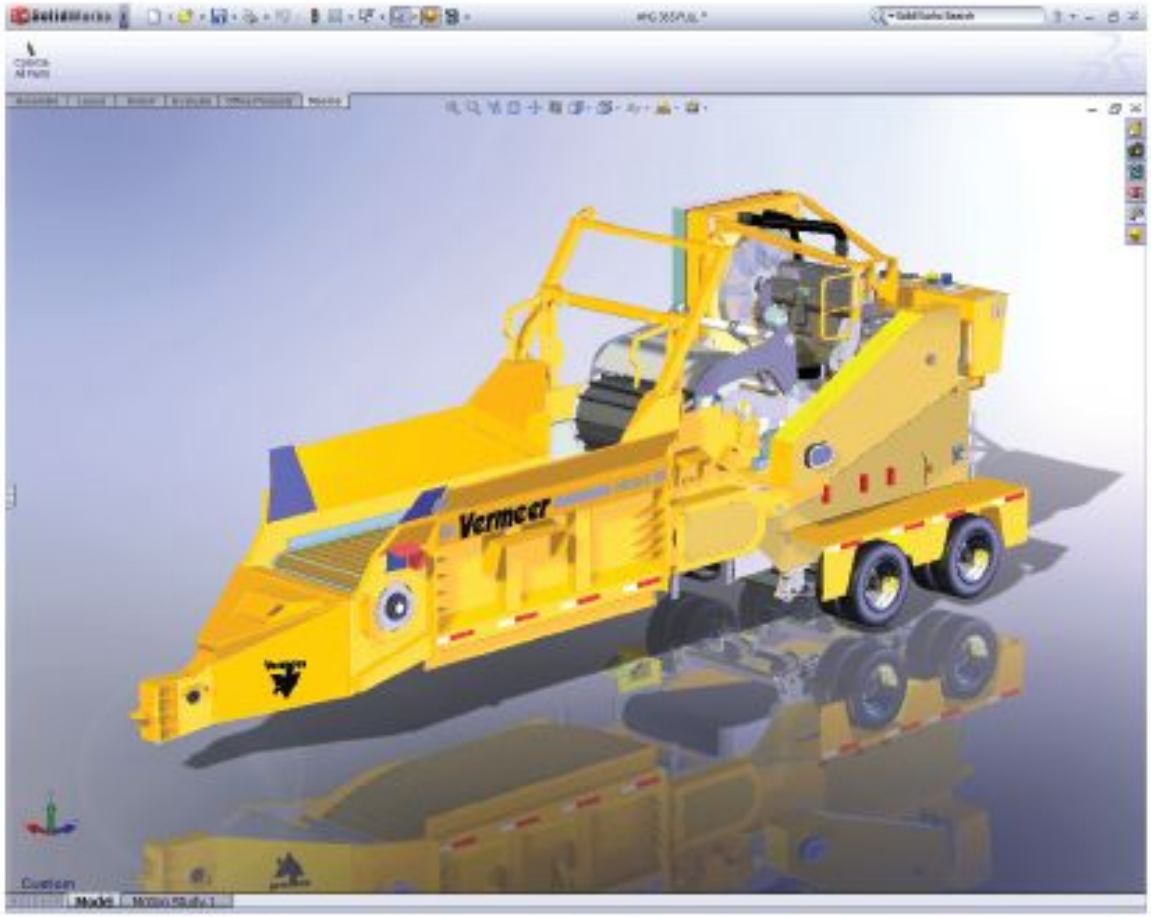
Reduce errors with interference and collision checking Cont.

Fortunately, the SolidWorks software has a solution for ***collisions*** also. In SolidWorks software you can ***"move"*** your design through its ***full range of motion*** while *continuously checking for collisions between parts*.

When a ***collision*** is detected, the ***motion will automatically stop***, and the ***interference*** will be ***highlighted***.

Reduce errors with interference and collision checking Cont.

SolidWorks software even provides an ***audible sound*** to let you know *when parts interfere*. If you make designs with *internal moving components*, ***Collision Detection*** is an *invaluable tool* to check the function of your design.



Vermeer uses SolidWorks software to check for interferences and collisions in large assembly designs, which range from 500 to 4,000 parts.

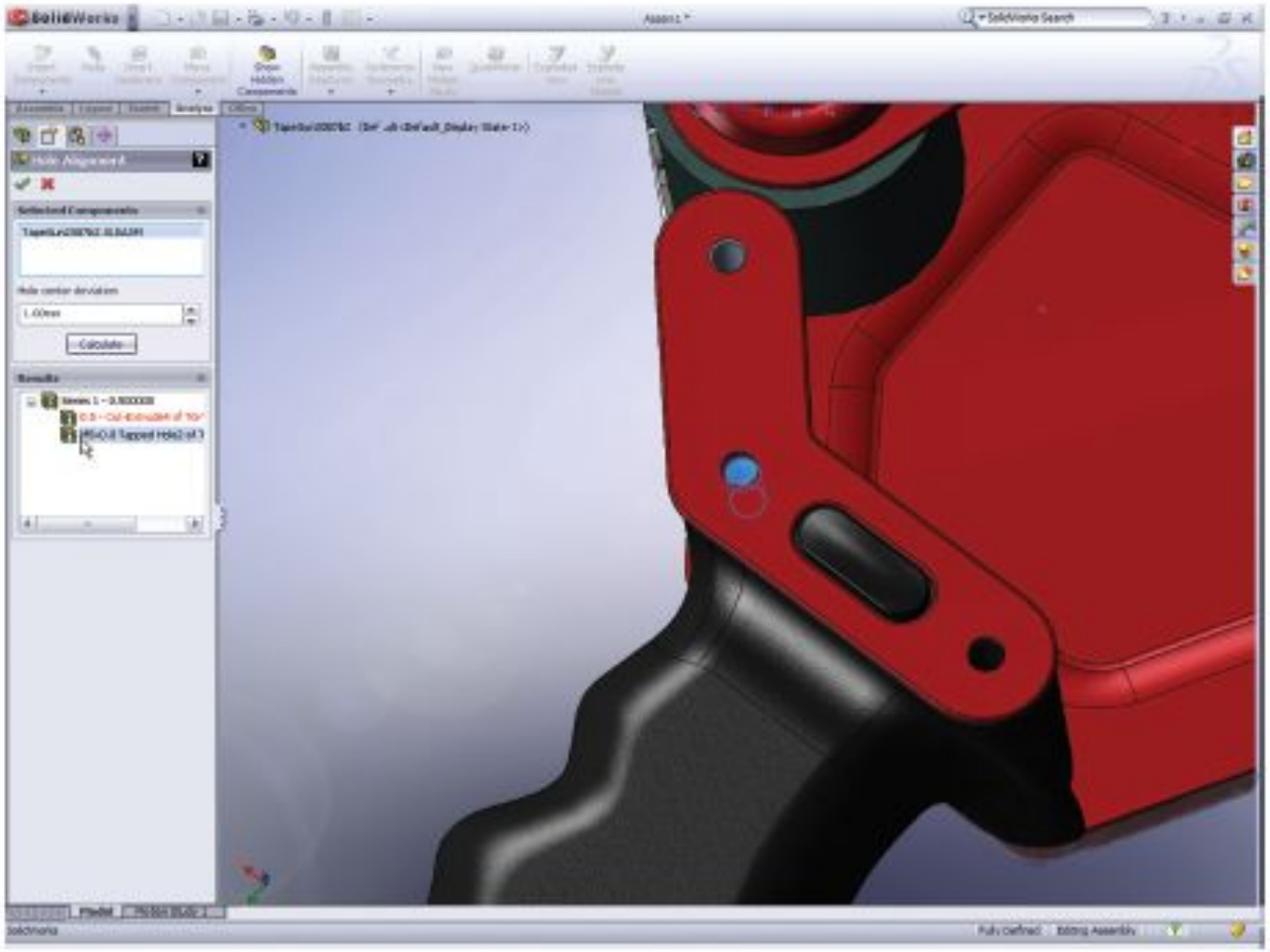
Reduce errors with interference and collision checking Cont.

Interferences also result from ***tolerancing*** *problems*. SolidWorks software provides the ability to ***automatically*** check *maximum and minimum tolerance conditions* with its ***TolAnalyst™*** functionality, so you can ensure the ***tolerances*** you are applying to your parts are appropriate.

Reduce errors with interference and collision checking Cont.

It even identifies which tolerances are the largest contributors to the tolerance stack-up problem, thereby saving you time in trying to figure out which tolerances to tighten, or which dimensioning schemes to change.

When ***fit*** and ***function*** ***errors go down***, ***efficiency*** ***goes up***, and you ***reduce time, labour, and material costs***.



SolidWorks software checks hole alignment automatically.

Case Study:

Wayne Tiffany

Automatic Systems Inc. (ASI) in Kansas City, Missouri

Automatic Systems Inc. (ASI) in Kansas City, Missouri, manufactures a wide range of industrial *material-handling equipment to move everything from coal to car bodies*.

Wayne Tiffany, *senior machine designer, explains 'they often deal with large machine designs that require complicated material handling paths'. He states 'that there is* ***no*** *reasonable way he can validate that his designs will fit together and function properly if he is using* ***2D***'.

Catching and correcting interferences and collisions in 3D is invaluable. If these interference and collision problems were not picked up until final assembly and test, it could ***cost*** *the company* ***10 or 100 times*** *as much to fix the problem, and of course it would most likely impact delivery of the machine to the customer.*

*Ultimately, says **Tiffany**, "SolidWorks software **saves** our company **substantial money** and **time** by catching problems before they go to manufacturing—these are problems that we had difficulty finding before in 2D."*

Summary:

Reduce errors with interference and collision checking

1. *2D CAD - **part interferences** are difficult to find, especially when the design becomes large and complicated.*
2. *2D CAD - multiple designers - assembly design duties - **potential** for **interfering parts** almost becomes a certainty* – errors
3. 3D CAD - *Interference* checking is ***automatic*** - ***every part*** can be checked to see if it ***interferes*** with any other part
4. *2D CAD - **no** practical way to check for a **collision***
5. *3D CAD - can **"move"** your design through its **full range of motion** while continuously checking for **collisions** between parts*

Task 3

Assemblies of 3D CAD models can be used to check for part *interference* and *collision – errors are discovered during the design rather than manufacturing process.*

1. What do ***YOU*** think they mean when they talk about *interference and collision detection* within an assembled product???
2. Why do ***YOU*** think it is difficult to detect interference between assembled parts when drawn in 2D???
3. How does ***Tiffany*** think a 3D CAD model which automatically detects *interference and collisions* between assembled parts ***benefit*** his ***design/ manufacturing organisation***???



4. Reuse existing designs

Reuse existing designs

Two *unique aspects* of SolidWorks software allow you to *make easy and extensive* ***reuse*** *of existing designs:* ***associativity*** and ***modifiability***.

- ***Associativity*** - means when you change a design model - the change ***automatically*** ripples through to all the other places where that model is used - the drawings, higher level assemblies, and more.

Reuse existing designs Cont.

- ***Modifiability*** - means you can change a part by clicking and changing a dimension and all other geometry on that part resizes appropriately and automatically.

For example, *if you have modelled a 125mm long plate with holes dimensioned **25mm** from either end, and then click on the 125mm dimension and change it to 250mm, the hole location would update, but the holes would still be **25mm** from either end of the part.*

Reuse existing designs Cont.

As you can see from this example, SolidWorks software also allows you to capture your design ***intent***, another very important aspect of designing in 3D.

The design ***"intent"*** in this example is that you always want the holes to be ***25mm*** *from the ends of the part no matter how long the part is*.



With SolidWorks software you can speed the creation of multiple configurations.

Reuse existing designs Cont.

Associativity and ***modifiability*** let you ***reuse existing designs*** to *create new versions or configurations easily*. You can readily create multiple new configurations of a single part by varying particular dimensions and features.

For example, you could make ***one socket head cap screw*** and then make every ***length and diameter variation*** from that ***one*** *design* just by ***changing dimensions***.

Reuse existing designs Cont.

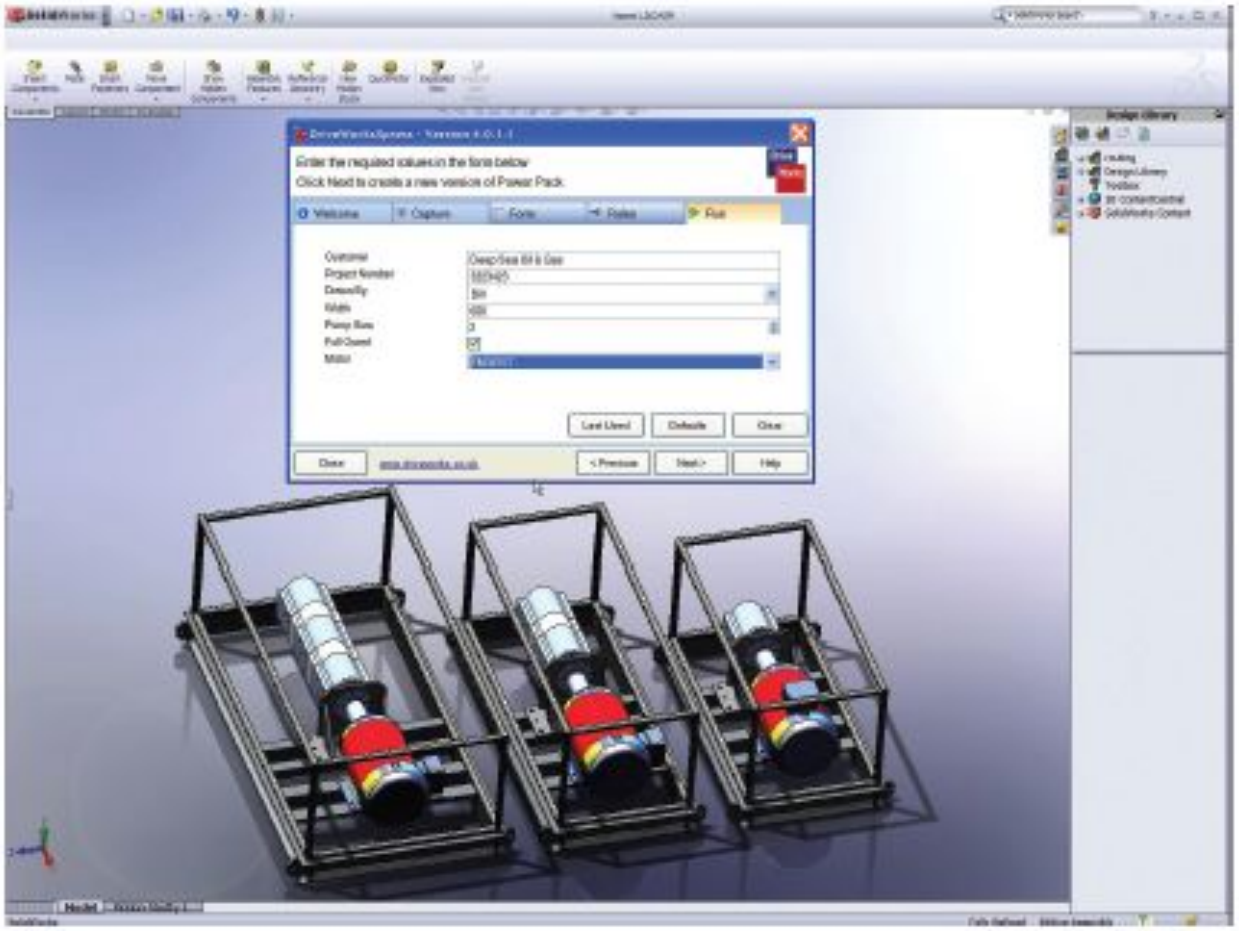
Likewise, you could make an entire catalogue of screws from one part by also *varying thread pitch, head type, and material as well as length and diameter*.

Reuse existing designs Cont.

SolidWorks software provides 3D parts catalogues containing supplier-certified 3D models both in the software and online at www.3DContentCentral.com

Case Study:

ASI's Tiffany notes that *he makes extensive use of the online parts catalogue in the design of his material-handling equipment*.



You can automate the creation of new designs with DriveWorksXpress.

Summary:
Reuse existing designs

1. ***Associativity*** *- means when you change a design model - the change* ***automatically*** *ripples through to all the other places where that model is used - the drawings, higher level assemblies*

2. ***Modifiability*** - means you can change a part by clicking and changing a dimension and all other geometry on that part resizes appropriately and automatically

3. ***Associativity*** and ***modifiability*** let you ***reuse existing designs*** to *create new versions or configurations easily – E.g. Socket head cap screws*

Task 4

The SolidWorks software enables the designer/drafter to ***reuse*** *existing designs* to create new versions or part configurations.

1. This section introduces the word ***'associativity'*** – what do you think this means???
2. This section introduces the word ***'modifiability'*** – what do you think this means???
3. How does ***Tiffany*** think the ability to ***reuse*** *an existing* 3D CAD model ***benefits*** his ***design/ manufacturing organisation***???



5. Accelerate development cycles with virtual testing and optimisation

Accelerate development cycles with virtual testing and optimisation

Speeding up a design cycle depends on more factors than simply streamlining the initial part or assembly design. Another ***major benefit*** with ***3D modelling*** is the insight it offers through ***virtual testing, analysis,*** and ***optimisation***, which take many forms.

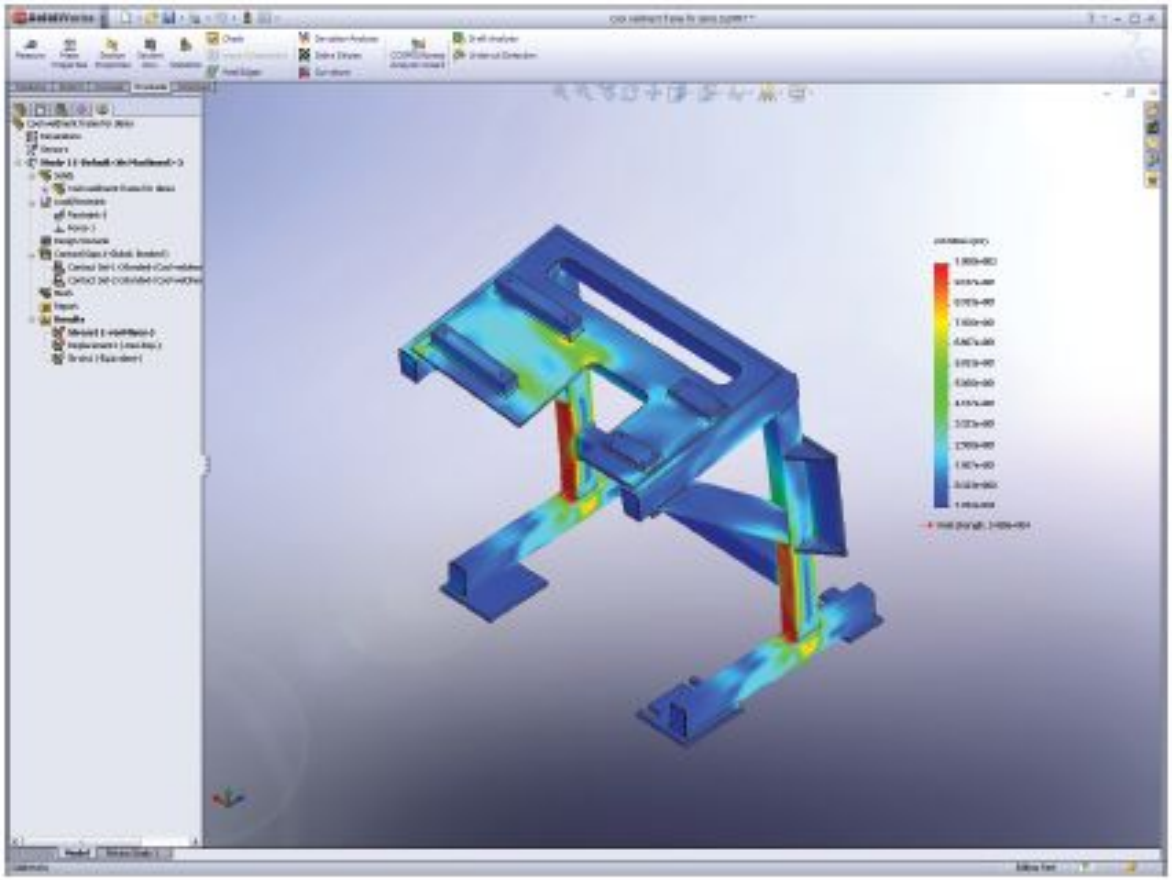
Case Study:

Wayne Tiffany

Automatic Systems Inc.
(ASI) in Kansas City, Missouri

Tiffany appreciates that working in SolidWorks software allows him to ***apply motion*** to the *parts of an assembly and quickly evaluate many different designs against operational requirements*.

Designers can assign a material type to a part and identify the ***mass properties***, including ***weight*** and ***centre of gravity***.



SolidWorks software provides an array of simulation and optimization tools to help you calculate forces due to motion, part stress, and deflection as well as vibration, flow, and effects of temperature.

Accelerate development cycles with virtual testing and optimisation Cont.

In addition the ***motion simulation*** capability found in SolidWorks software allows you to ***evaluate*** the effects of *different motor performance curves, friction, springs, gravity, and other physical characteristics of a design*.

So rather than just running an animation of the machine in motion, you can simulate what really is happening in your design.

Accelerate development cycles with virtual testing and optimisation Cont.

The ***motion simulation*** *automatically* ***calculates forces*** *on critical components, like bearings, bushings, and linkages*. This information is then used to ***calculate*** *part strengths, deflection, fatigue, and safety factors*.

Thermal, vibration, and flow analysis are also provided directly inside SolidWorks software so that your design can be optimised.

Accelerate development cycles with virtual testing and optimisation Cont.

In this way, SolidWorks software can be used to solve typical design problems, such as:

- *How much weight can I remove from this part yet still have it function within a certain factor of safety?*
- Will the part bend too much if it is made from a different material?
- In an electronic enclosure, will a specific component overheat from lack of airflow?

Summary:

Accelerate development cycles with virtual testing and optimisation

1. *3D CAD - major benefit* – speeds up the ***design process*** - ***virtual testing, analysis,*** and ***optimisation*** – *saves* ***time*** *and* ***money*** *as the* ***working conditions*** *of components and products can be* ***simulated*** *using software – reducing the need to produce* ***physical prototypes*** *for* ***testing***

2. *3D CAD - these software tools include – Finite Element Analysis (FEA) – analysis of component stresses and strains; Thermal, Fluid and Vibration Analysis; Motion Simulation - calculation of forces; Application of Material Properties to components*

Task 5

The SolidWorks software enables the designer/drafter to accelerate the design/development cycles of parts and products with *virtual testing and optimisation.*

1. **FEA** means **Finite Element Analysis** – this software would enable the designer/drafter to do what with a 3D CAD model???
2. How does ***Tiffany*** think the ability to ***analyse*** a *3D CAD model* using special computer software ***before*** the part or product is ***physically*** *manufactured* ***benefits*** his ***design/ manufacturing organisation***???

6. Design for manufacturing

Design for manufacturing

Many of the new manufacturing technologies rely on the availability of a 3D CAD model as a starting point.
For example, ***three-axis (plus) NC programming, rapid prototyping, mould design,*** and even ***sheet-metal manufacturing*** now require 3D models that can be referenced to create *NC toolpaths, SLA models (Stereolithography - 3D Printed models), and sheet metal flat patterns* with proper bend allowances.

Design for manufacturing

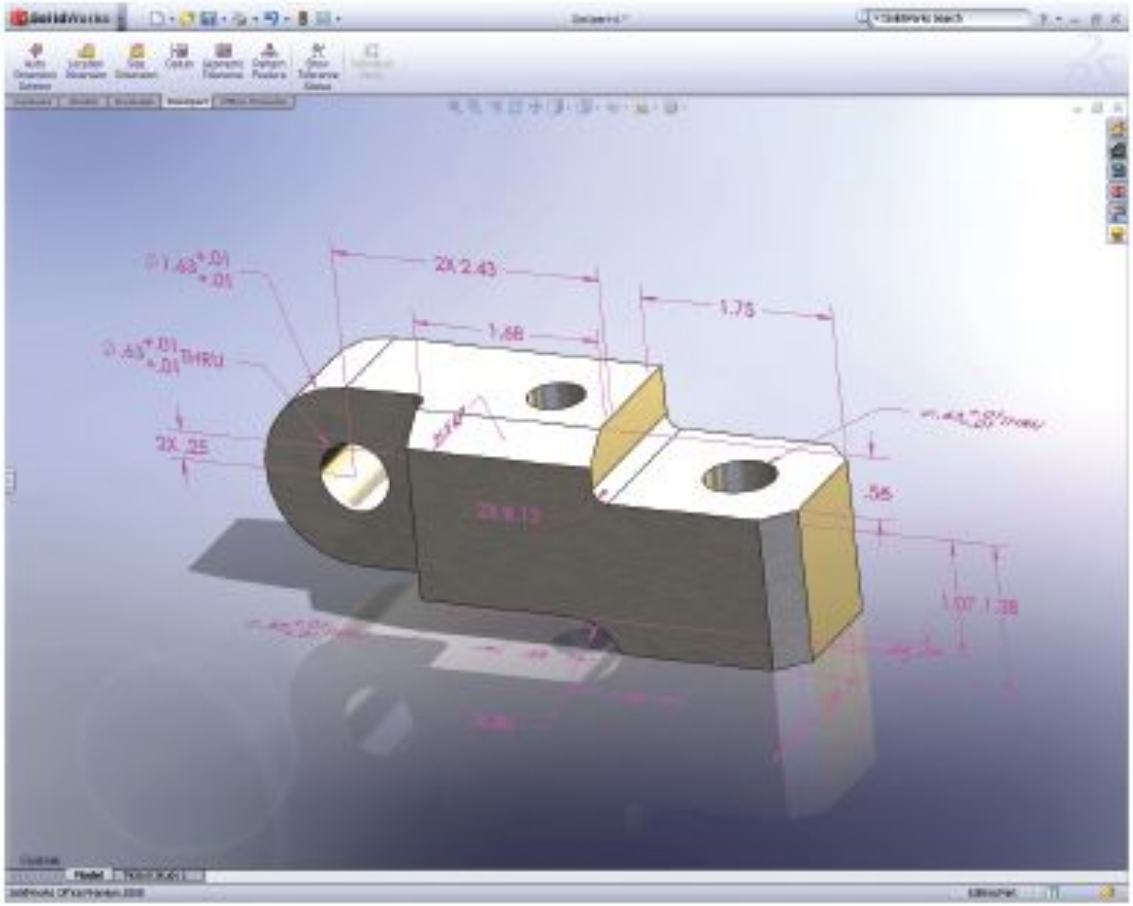
More and more companies are turning to ***rapid prototyping*** and ***manufacturing*** to *reduce the turnaround time for both prototype parts and customised final parts*, such as hearing aid shells.

These rapid systems operate from 3D modelling input generated in ***standard template library*** **(STL) format -** a file format that you can output from your 3D CAD model by simply saving your model in that format.

Design for manufacturing Cont.

In many cases, 2D drawings are not even required for manufacturing.

For example, SolidWorks software can ***output*** the 3D model complete with detailed ***dimensioning*** and ***tolerancing*** information as well as notes. In this way, all the data and notes needed to manufacture the part are included directly in the 3D CAD model.



With SolidWorks software you can automatically dimension 3D models according to industry standards.

Design for manufacturing Cont.

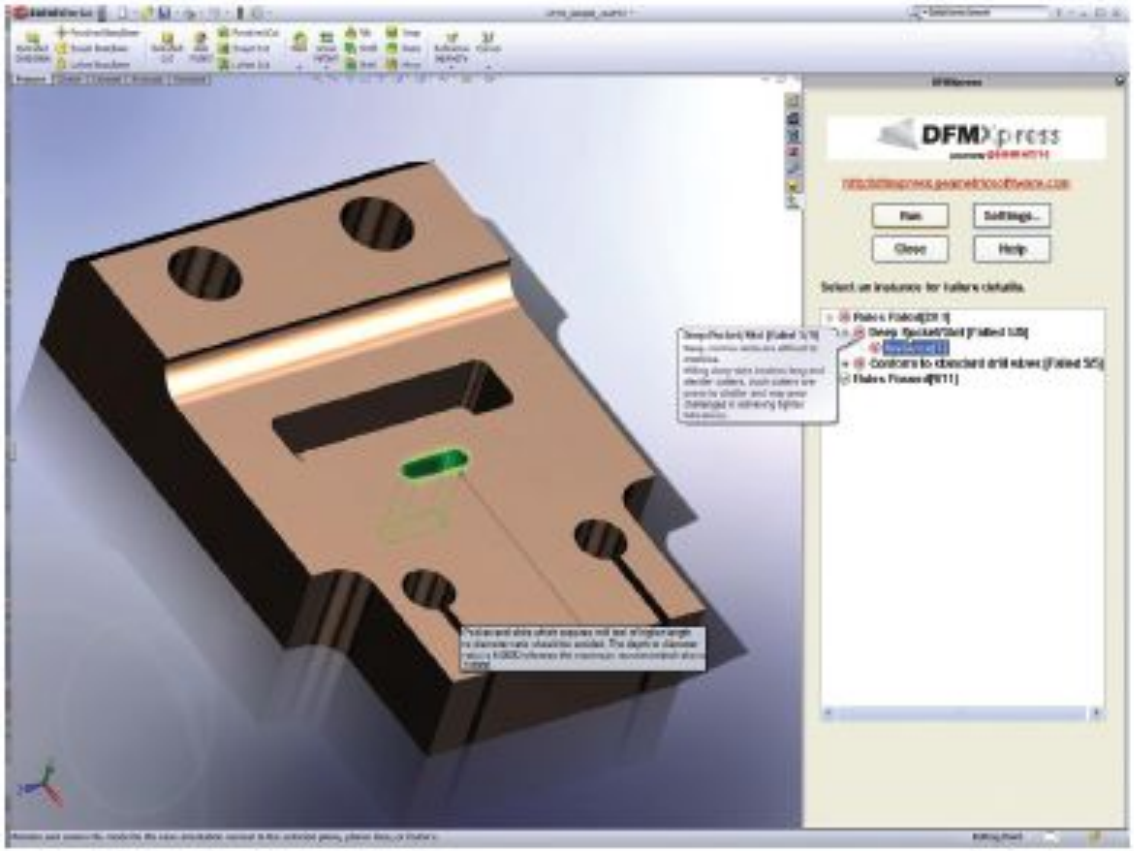
Communication between *manufacturing and engineering* is also greatly improved through the use of 3D.

There is a wide range of 3D digitising and inspection equipment that also requires 3D CAD models. With the complete details at hand in the 3D CAD model, you can more easily *verify moulded, forged, stamped, and cast parts, including the details of undercuts, draft, thickness, and surface topology*.

Design for manufacturing Cont.

With SolidWorks software, the manufacturing department is provided with more useful and richer information:

- ***Machinists*** can *measure and section 3D parts live at the touch of a button*.
- ***Assemblers*** and ***fabricators*** can *explode assemblies, and zoom in and hide parts from view to see details they could never see before in 2D drawings*.



With DFMXpress you can review more closely the features of your designs that are costly and difficult to machine.

Summary:

Design for manufacturing

1. Modern manufacturing technologies require a 3D CAD model as a starting point – e.g. ***three-axis (plus) NC programming, rapid prototyping, mould design,*** even ***sheet-metal manufacturing*** and ***inspection equipment*** *(CMM – Coordinate Measure Machine)*
2. 2D drawings are not always required for manufacturing – 3D CAD software can ***output*** the 3D model complete with detailed ***dimensioning*** and ***tolerancing***

Task 6

Many of the new manufacturing technologies rely on the availability of a 3D CAD model as a starting point.

1. 3D CAD models can be used with other forms of manufacturing technologies – name ***three*** of these technologies.
2. The 3D CAD model often has to be saved as an ***SLT file*** – what does this file extension stand for???
3. How do ***YOU*** think the ability to convert a 3D CAD model so it can be used with another form of manufacturing technology would ***benefit*** a ***design/manufacturing organisation***???

7. Create bills of materials and manage data smoothly

Create bills of materials and manage data smoothly

Working with an associative 3D CAD system also guarantees an ***accurate*** *and* ***current bill of materials (BOM)***.

The BOM is always accurate because it ***automatically updates*** with changes you make to *parts and assemblies*.

Create bills of materials and manage data smoothly Cont.

Data management is always critical to the product development process.

Designs often go through *several stages of development*, such as *concept, detailed engineering design, assembly and test, and final release to production*.

Create bills of materials and manage data smoothly Cont.

Controlling data is especially important when you consider all the personnel that are required to develop and manufacture a product: *engineering, manufacturing, purchasing, sales, marketing, vendors, and even customers* need to know that they are looking at the *correct, most current data*.

Create bills of materials and manage data smoothly Cont.

SolidWorks ***product data management*** *software* ensures that the proper release procedures and approvals for designs are executed from stage to stage.

8. Sales and marketing tools

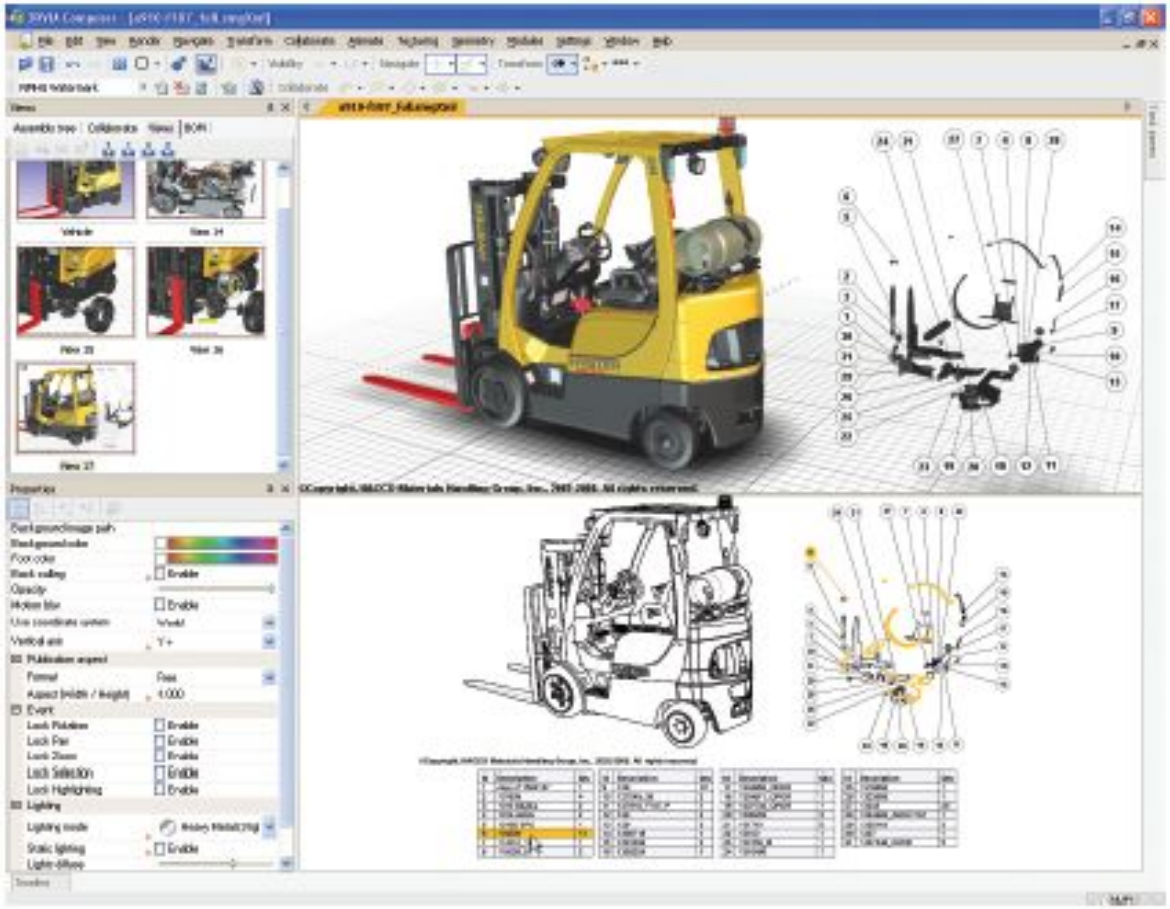
Sales and marketing tools

Sales and marketing can also reap the benefits of 3D CAD data. ***Publication tools*** fully support the use of 3D CAD data to allow the *creation* of ***photorealistic 2D images, 3D models, and animations*** that appeal more to *customers and consumers*.

Look at any major car manufacturer's website, and you will see the demand to present products in a *more complete, colourful, and photorealistic way*.

Sales and marketing tools Cont.

Photorealistic models and ***animation*** *capabilities* as well as *rapid prototyping* also allow ***marketing*** to perform *product market research* at much *lower costs* than *actually designing and producing prototypes* in the conventional manner.



Use 3D CAD models to more effectively communicate your designs on the web, in animations, and more.

Conclusion

3D CAD design has many benefits over working in 2D:

- *Product visualisation and presentation are improved*
- *Parts and drawing views update automatically and accurately*
- *Interference and collision checking provides an automated, error-free way to check for interferences and collisions before manufacturing the product.*

Two facts are clear from this presentation:

1. The world of design and manufacturing is definitely moving to 3D

2. Customers and designers are all demanding 3D to enhance ***design*** and ***communication***.

Organic shapes, found in the design of consumer products, from cars to mobile phones, are much easier to model and manufacture in 3D.

And with the ***tolerances*** expected from today's consumer product and automotive designs, ***2D sections*** are now ***unacceptable*** for defining these ***organic shapes*** because 2D does not provide enough surface control between sections.

Finally, 3D modelling breathes new life into the design process, it attracts and retains designers and engineers eager to use the latest design tools, and even improves a companies image to customers and vendors.

It is a rare business today that can function productively without email and websites - 3D design is another big step to operating fully in this increasingly digital world.